Project Idea

Topic: A.N.T – Academic Network & Talent Map Platform

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Idea of Project & Background

Universities and organizations have different types of information about their people, such as academic programs, courses, projects, certifications, internship experience, research, publications, and skills. In higher education, these data may come from different databases or systems, which can make the information more difficult to access, understand, and use for decision-making [1].

This can create difficulties when trying to understand the expertise and relationships within an organization. It can also make it difficult to identify suitable researchers or collaborators. Previous studies have shown that finding suitable university researchers is one of the challenges in university-industry collaboration, especially when information about researchers and their expertise is difficult to identify [2]. Research networking systems have also been studied as a way to help researchers find collaborators with suitable expertise [3].

To reduce these problems, A.N.T. (Academic Network & Talent Map Platform) is proposed to connect and visualize relationships between people, skills, courses, projects, research, publications, and job roles. The platform will use a combination of data visualizations, with an interactive Network Graph using a Force-Directed Layout as the main visualization. Force-directed graphs have been used to visualize academic networks and help users explore connections between related academic information [4]. The layout can also help show network structure by placing connected nodes based on their relationships [5].

For example, A.N.T. can show people related to cybersecurity together with their courses, projects, research, certifications, and relevant job roles such as SOC Analyst. This allows stakeholders to see who has related skills, what information supports those skills, and how people, academic activities, and job requirements are connected.

KMUTT will be used as a case study for the project. The platform will be designed so that it can later be adapted for other universities or organizations.

Objective

- To clearly represent talent and skills based on supporting evidence such as courses, projects, research, publications, certifications, and work or internship experience.
- To allow stakeholders to understand the relationships between people, skills, academic activities, research, and job roles.
- To help stakeholders identify people with related expertise and understand what information supports their skills.

- To compare existing skills with skills required for specific job roles and identify areas that may need further development

Identified Problems

- Information about people, academic activities, skills, and research is presented separately, making relationships between the data difficult to see.
- University executives may not have a clear overview of expertise available within the university.
- Information barriers can make it difficult to identify possible academic and research collaborations.
- External stakeholders may have difficulty finding suitable experts for research, funding, or collaboration.
- Students may have difficulty finding information about university partnerships and relevant academic contacts.

Scope of Work

- Identify the data needed to represent people, skills, courses, projects, research, publications, and job roles, including the relationships between them.
- Study relevant KMUTT and external data sources, including their availability, access, and limitations.
- Design an interactive Network Graph using a Force-Directed Layout to show relationships between people, skills, academic information, research, and industry job requirements.
- Develop the A.N.T. platform, including the database, frontend, backend, search, filtering, and interactive data visualization, with KMUTT as a case study.
- Evaluate whether the platform can clearly show talent, supporting evidence for skills, and relationships between academic information and job requirements, and identify limitations of the platform.

Expected Results

- A working A.N.T. platform that visualizes relationships between people, skills, courses, projects, research, publications, and job roles.
- An interactive Network Graph using a Force-Directed Layout that allows users to explore related people, skills, and supporting evidence.
- A clearer view of talent and skill relationships within KMUTT, including connections to relevant industry job requirements.
- Evaluation results showing whether users can understand the relationships and information presented by the platform.
- Identified limitations of the data and platform that can be used for further improvement or adaptation to other organizations.

Reference

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